RESEARCH PROJECT PRESENTATION & VERIFICATION AUDIT

Reliability-Aware Multimodal Analysis for Parkinson’s Disease Detection

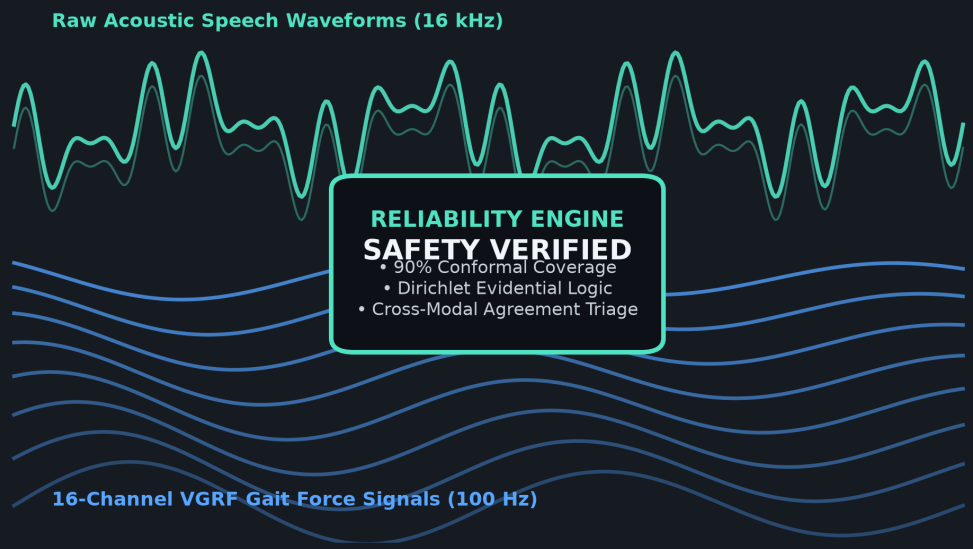
Objective Biomechanical Gait & Acoustic Speech Analysis Under Epistemic Uncertainty

• **Dual Clinical Modalities:** 16-channel Vertical Ground Reaction Force (VGRF) biomechanics from PhysioNet (165 subjects) and clinical acoustic speech recordings from MDVR-KCL (37 subjects).

• **Beyond Standard Classification:** Incorporating Dirichlet evidential neural heads, Deep Ensembles (M=5), Temperature Scaling, and Split Conformal Prediction (90% coverage guarantees).

• **Controlled 1D CNN Baselines:** Scientific benchmarking against fair raw-input 1D CNN models on identical subject partitions to isolate the value of temporal BiLSTM and Wav2Vec2 representations.

• **Research Prototype Framing:** Developed as an academic decision-support framework to flag clinical uncertainty; not intended as an autonomous medical diagnostic device.



CORE SYSTEM ARCHITECTURE HIGHLIGHTS

1. Gait Biomechanics: 1D CNN + BiLSTM on 16 force channels (500 time steps @ 100 Hz).

2. Speech Audio: Pre-trained Wav2Vec2 fine-tuned on clinical speech tasks.

3. Reliability Layer: Closed-form Dirichlet uncertainty (u = K / S) and 90% conformal sets.

4. Conflict Policy: Decision-level fusion that detects and flags cross-modal disagreements.

The Reliability Gap in Medical Machine Learning

Why conventional softmax classifiers fail in clinical neurodegenerative screening

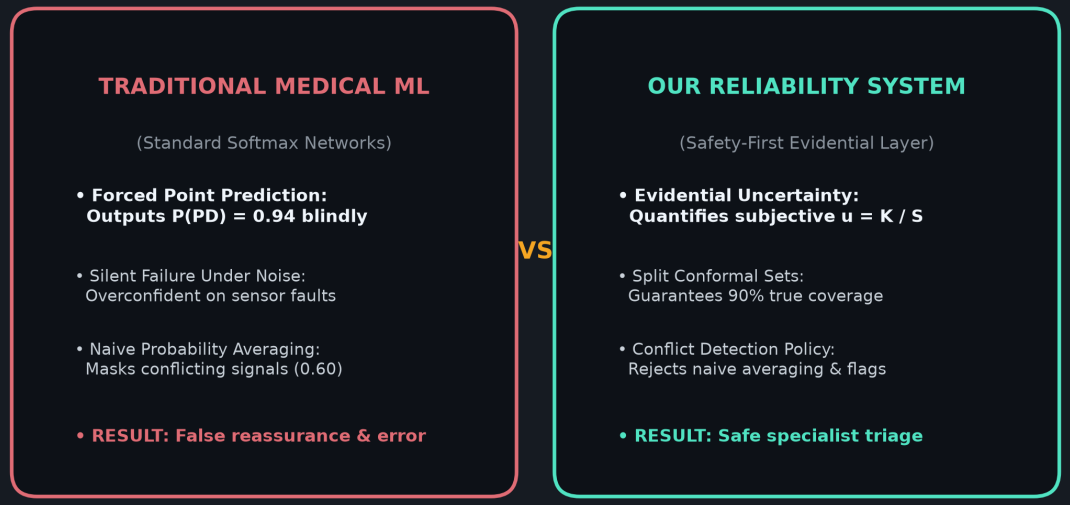
CLINICAL & STATISTICAL FAILURE MODES

1. Clinical Nuance & Symptom Overlap:
Parkinsonian motor degradation causes bilateral gait asymmetry and subtle vocal hypophonia/dysarthria that overlap significantly with normal healthy aging, requiring high sensitivity without excessive false alarms.

2. The Silent Overconfidence Failure Mode:
Standard neural networks trained with cross-entropy force uncalibrated point probabilities, outputting >95% confidence on out-of-distribution inputs and corrupted sensor streams without flagging doubt.

3. Inter-Cohort Distribution Shift:
Sensor shoe designs, floor plates, acoustic hardware, and speaking protocols vary across clinical sites, causing models that look accurate on one hospital cohort to degrade sharply on unseen clinics.

4. The Hazard of Naive Multimodal Fusion:
Averaging conflicting predictions (e.g., Gait P(PD)=0.85 vs Voice P(PD)=0.35) yields a false middle ground (0.60) that silently obscures severe sensor or task conflict.



OUR SAFETY-FIRST TRIAGE SOLUTION

• **Evidential Epistemic Heads:** Parameterize Dirichlet distributions Dir(α_k) to quantify subjective uncertainty in a single forward pass without expensive sampling.

• **Finite-Sample Conformal Sets:** Outputs statistical sets ({PD}, {HC}, or {HC, PD}) guaranteeing 90% true-label coverage, converting forced guesses into actionable clinical ambiguity flags.

• **Conflict Detection Rules:** Explicitly triggers a MODALITY_CONFLICT alert when gait and voice diverge, safely directing the case to specialist human review.

Project Objectives & Technical Contributions

Five concrete engineering contributions implemented and verified in the repository

1. Temporal Biomechanics (Gait)

End-to-End 1D CNN + BiLSTM

Developed an end-to-end architecture on 16 continuous force channels to capture both intra-stride pressure impulses and long-range inter-stride bilateral asymmetry across 500-step windows.

2. Speech Representation (Voice)

Fine-Tuned Wav2Vec2 Audio

Fine-tuned pre-trained self-supervised Wav2Vec2 representations on clinical speech tasks, benchmarked against a newly developed raw-waveform 1D CNN baseline on identical splits.

3. Evidential Uncertainty Logic

Subjective Logic Dirichlet Heads

Implemented evidential neural heads parameterizing Dirichlet distributions. Delivers non-negative evidence and closed-form epistemic uncertainty ($u = K / S$) in a single deterministic pass.

4. Cross-Site Generalization & Sensor Corruption Benchmarking

Leave-One-Cohort-Out (LOCO) Robustness

Evaluated models across three independent hospital sites (165 subjects) and stress-tested calibration stability (Expected Calibration Error) under Gaussian sensor noise, channel dropout, and force signal attenuation.

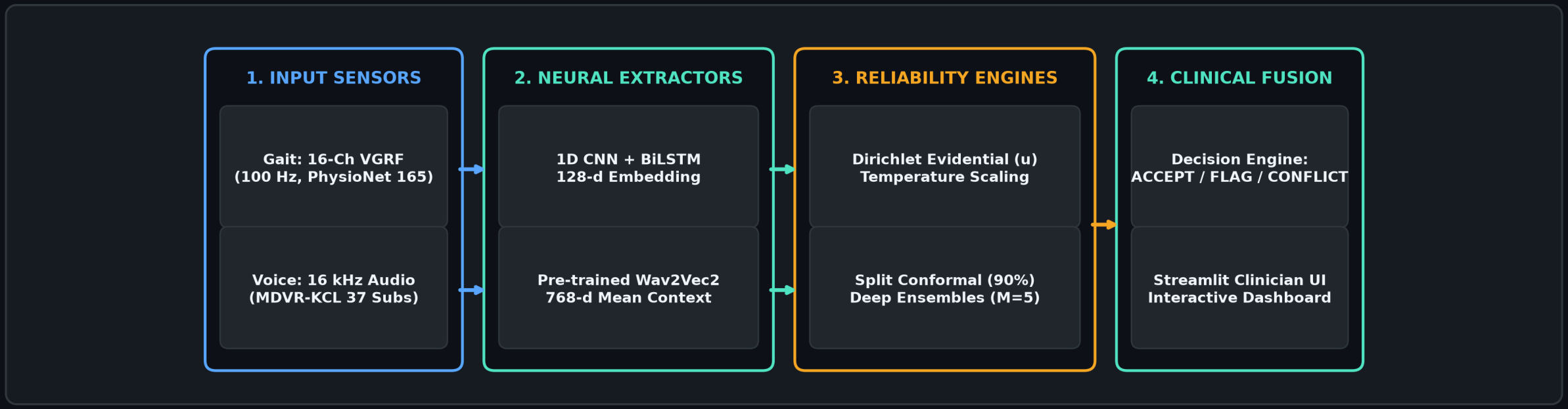
5. Safety-First Cross-Modal Fusion & Clinician Dashboard

Decision-Level Agreement Policy & Live UI

Engineered a decision-level reliability fusion engine that outputs ACCEPT, FLAG, or CONFLICT statuses, integrated into an interactive Streamlit demonstration with live transparency panels.

End-to-End Reliability System Architecture

Complete dataflow from raw biomechanics & acoustic signals to reliability triage



1. DUAL FEATURE EXTRACTORS

• **Gait Stream:** 16-channel VGRF (100 Hz) → 5.0 s windows → 1D CNN + BiLSTM → 128-d dense embedding.

• **Voice Stream:** 16 kHz Mono → 3.0 s chunks → Wav2Vec2 Contextual Transformer → 768-d mean vector.

2. RELIABILITY VERIFICATION

• **Dirichlet Evidential:** Single-pass subjective uncertainty u = K / S.

• **Conformal Prediction:** Split conformal sets guaranteeing 90% true coverage.

• **Deep Ensembles:** M=5 independent seeds capturing epistemic variance.

3. DECISION-LEVEL FUSION

• **High Concordance:** Singleton overlap → **ACCEPT**.

• **Ambiguous Bounds:** High entropy/variance → **FLAG**.

• **Cross-Modal Conflict:** Opposing predictions → **CONFLICT**.

• **Delivery:** Interactive Streamlit clinician dashboard.

Clinical Datasets & Zero-Leakage Preprocessing Protocol

PhysioNet (165 subjects) and MDVR-KCL (37 subjects) under strict patient-level separation

CLINICAL DATASET SPECIFICATIONS

Gait Dataset (PhysioNet - Gait in Parkinson's Disease):
- **Total Cohort:** 165 subjects (93 PD, 72 Healthy Controls)
- **Clinical Sites:** Galit (Ga: 47), Hausdorff (Ju: 54), Silberstein (Si: 64)
- **Sensor Setup:** 16 continuous Vertical Ground Reaction Force (VGRF) sensors (8 Left foot, 8 Right foot) sampled at 100 Hz.
- **Temporal Windowing:** 5.0 s duration (500 time steps) with 50% overlap (2.5 s step).
- **Evaluated Windows:** 14,440 continuous biomechanical samples.

Voice Dataset (MDVR-KCL - King's College London):
- **Total Cohort:** 37 subjects (22 PD, 15 Healthy Controls)
- **Acoustic Tasks:** ReadText and SpontaneousDialogue recordings.
- **Audio Format:** 16 kHz single-channel float32 waveforms.
- **Segmentation:** Overlapping temporal chunks aggregated to patient level.

ZERO-LEAKAGE SCIENTIFIC GUARANTEE

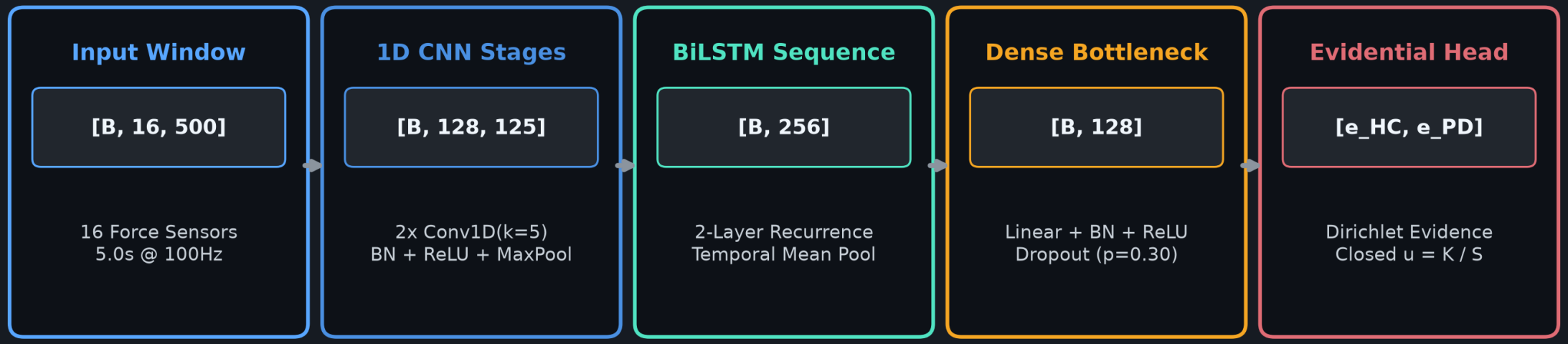
Strict Subject-Level Stratification (Seed 42):
- **Zero Patient Overlap:** No individual appears in more than one partition:
  $\text{Train} \cap \text{Val} = \emptyset$, $\text{Train} \cap \text{Test} = \emptyset$, $\text{Val} \cap \text{Test} = \emptyset$.
- **Gait Partitions:** 115 Train, 25 Validation, 25 Held-Out Test subjects (11 HC, 14 PD).
- **Voice Partitions:** 26 Train, 5 Validation, 6 Held-Out Test subjects (3 HC, 3 PD).

Strict Train-Only Normalizer Estimation:
- Channel-wise mean and standard deviation for Z-score normalization are computed strictly on training subjects.
- Validation and test subjects are normalized using frozen training parameters.

Elimination of Contamination:
Completely prevents window-level and recording-level leakage where overlapping windows from the same patient pollute both train and test sets—a rampant flaw in published literature.

Gait Biomechanical Modeling (1D CNN + BiLSTM)

Capturing local pressure impulses and bilateral inter-stride temporal phase transitions



STAGE 1: 1D CONVOLUTIONS

- **Input Tensor:** [B, 16, 500] (16 force sensors, 500 time steps).
- **Block 1:** Conv1D(16→64, k=5, p=2) → BN → ReLU → MaxPool(2) → [B, 64, 250].
- **Block 2:** Conv1D(64→128, k=5, p=2) → BN → ReLU → MaxPool(2) → [B, 128, 125].
- **Role:** Extracts local pressure impulses across foot sensors.

STAGE 2: BIDIRECTIONAL LSTM

- **Transpose:** [B, 125, 128] (Sequence Length, Features).
- **2-Layer BiLSTM:** Hidden size 128 per direction → [B, 125, 256].
- **Global Pooling:** Temporal mean pooling → [B, 256].
- **Critical Role:** Tracks long-range bilateral stride phase transitions and timing degradation.

STAGE 3: DENSE & EVIDENTIAL

- **Dense Bottleneck:** Linear(256→128) → BN → ReLU → Dropout(0.30).
- **BCE Head:** Linear(128→1) + Sigmoid.
- **Evidential Head:** Linear(128→2) + Softplus → Non-negative evidence [e_HC, e_PD] ≥ 0.
- **Closed Uncertainty:** u = 2 / (e_HC + e_PD + 2).

Speech Representation Learning (Wav2Vec2)

Self-supervised acoustic representation learning fine-tuned on clinical vocal recordings



1. PRE-TRAINED BACKBONE

• **Pre-trained Corpus:** facebook/wav2vec2-base pre-trained on 100k+ hours of unlabelled speech.

• **7-Layer CNN Encoder:** Temporal convolution extracts latent acoustic tokens from raw 16 kHz waveforms.

• **12 Transformer Blocks:** Contextualizes acoustic representations over continuous speech.

2. DOWNSTREAM FINE-TUNING

• **Temporal Mean Pooling:** Averages contextual tokens across 3.0 s chunks into a 768-dim vector.

• **Classifier Head:** Linear(768→256) → ReLU → Dropout(0.30) → Linear(256→2).

• **Training Dynamics:** Fine-tuned with gradient accumulation (Batch=2, Accum=2, LR=10^-5) under Cross-Entropy.

3. SUBJECT-LEVEL INFERENCE

• **Chunk Level:** Generates posterior probabilities P(PD) on overlapping 3.0 s segments.

• **Recording Level:** Averages chunk posteriors across individual ReadText or SpontaneousDialogue tasks.

• **Patient Level:** Mean of task posteriors forms the final clinical probability score.

Safety & Reliability Mechanisms: 5-Pillar Framework

Multi-layered uncertainty quantification, probability calibration, and finite-sample guarantees

1. Dirichlet Evidential Networks

Subjective Logic & Closed-Form Uncertainty

Treats network outputs as evidence $e_k \geq 0$ parameterizing a Dirichlet distribution $Dir(\alpha_k)$. Closed-form epistemic uncertainty: $u = K / S$ is computed in a single pass without sampling.

2. Deep Ensembles (M = 5)

Multi-Seed Predictive Variance

Trains 5 independent neural networks initialized with distinct random seeds. Disagreement is quantified via predictive variance σ^2_{ens} and mutual information between member predictions.

3. Monte Carlo Dropout (T = 20)

Stochastic Bayesian Sampling

Preserves dropout masks ($p = 0.30$) during test-time inference across T=20 stochastic forward passes to approximate the Bayesian posterior distribution and model parameter uncertainty.

4. Temperature Scaling Probability Calibration

Post-Hoc Negative Log-Likelihood Optimization

Optimizes a single scalar parameter T^* on validation logits (z / T^*) to minimize Expected Calibration Error (ECE), eliminating systematic overconfidence without altering ROC-AUC rank ordering.

5. Split Conformal Prediction (1 - α = 0.90 Coverage)

Distribution-Free Finite-Sample Guarantees

Uses non-conformity scores $s_i = 1 - P(y_i|x_i)$ on calibration subjects to compute exact empirical quantiles. Guarantees 90% statistical coverage: singleton ({PD}), ambiguous ({HC, PD}), or empty sets.

Experimental Design & Controlled Baseline Framework

Establishing strict raw-input 1D CNN baselines and archiving literature 2D CNN references

THE FAIR COMPARISON MANDATE

Scientific Integrity Principle:
To scientifically isolate the benefit of BiLSTM recurrent sequence modeling and Wav2Vec2 self-supervised representations, baseline models must operate on the exact same raw inputs and identical subject-level data partitions.

Active Primary Baselines Created:
• **Gait 1D CNN Baseline:** Pure 3-stage Conv1D (16→64→128→256) + Global Average Pooling + Linear Classifier. Operates directly on raw 16-channel, 500-step VGRF windows (no LSTM, no evidential head).
• **Voice 1D CNN Baseline:** Pure 4-stage Conv1D (1→32→64→128→256) + Adaptive Pooling on raw 16 kHz waveforms (no pre-trained weights, no STFT).

Archived Literature 2D CNN Models:
• The paper-inspired models (Vásquez-Correa et al., 2019) used 2D CNNs on STFT representations.
• We preserved these in Gait/baselines/base_paper_cnn/ and Voice/baselines/base_paper_cnn/ as archived supplementary references; excluded from active comparisons.

CONTROLLED EVALUATION MATRIX

GAIT MODALITY BENCHMARK (165 Subjects):
• **Partitions:** 115 Train | 25 Validation | 25 Held-Out Test (11 HC, 14 PD)
• **Input Data:** Raw 16-channel continuous force signals (500 steps @ 100 Hz)
• **Baseline:** 1D CNN (Active Primary Benchmark)
• **Proposed Models:** 1D CNN + BiLSTM (BCE) & Dirichlet Evidential Model
• **Evaluated Metrics:** Accuracy, Balanced Acc, Sensitivity, Specificity, F1, ROC-AUC.

VOICE MODALITY BENCHMARK (37 Subjects):
• **Partitions:** 26 Train | 5 Validation | 6 Held-Out Test (3 HC, 3 PD)
• **Input Data:** Raw 16 kHz audio waveforms
• **Baseline:** 1D CNN Raw Waveform (Active Primary Benchmark)
• **Proposed Model:** Pre-trained Wav2Vec2 + Contextual Classification Head

AUDIT PROTOCOL GUARANTEE:
Identical random seed (42), frozen train normalizers, subject-level prediction aggregation, and 95% bootstrap confidence intervals computed using subject-level resampling.

Gait Evaluation: Baseline vs. Proposed Models

Independently recalculated subject-level metrics on 25 held-out test subjects (11 HC, 14 PD)

AUDITED GAIT BENCHMARK (N = 25)

1D CNN Baseline (Active Benchmark):
- Accuracy: **68.0%** | Balanced Accuracy: **68.51%**
- Sensitivity: **64.29%** | Specificity: **72.73%** | F1: **69.23%**
- ROC-AUC: **0.6818** | Brier: 0.2592 | NLL: 1.2645
- 95% Bootstrap CI: Acc [0.480, 0.880], AUC [0.462, 0.909]

Proposed 1D CNN + BiLSTM (BCE Loss):
- Accuracy: **72.0% (+4.0%)** | Balanced Acc: **71.10%**
- Sensitivity: **78.57% (+14.28% gain!)** | Specificity: **63.64%**
- ROC-AUC: **0.7532 (+0.0714 gain!)** | F1: **75.86%**
- 95% Bootstrap CI: Acc [0.520, 0.880], AUC [0.535, 0.942]

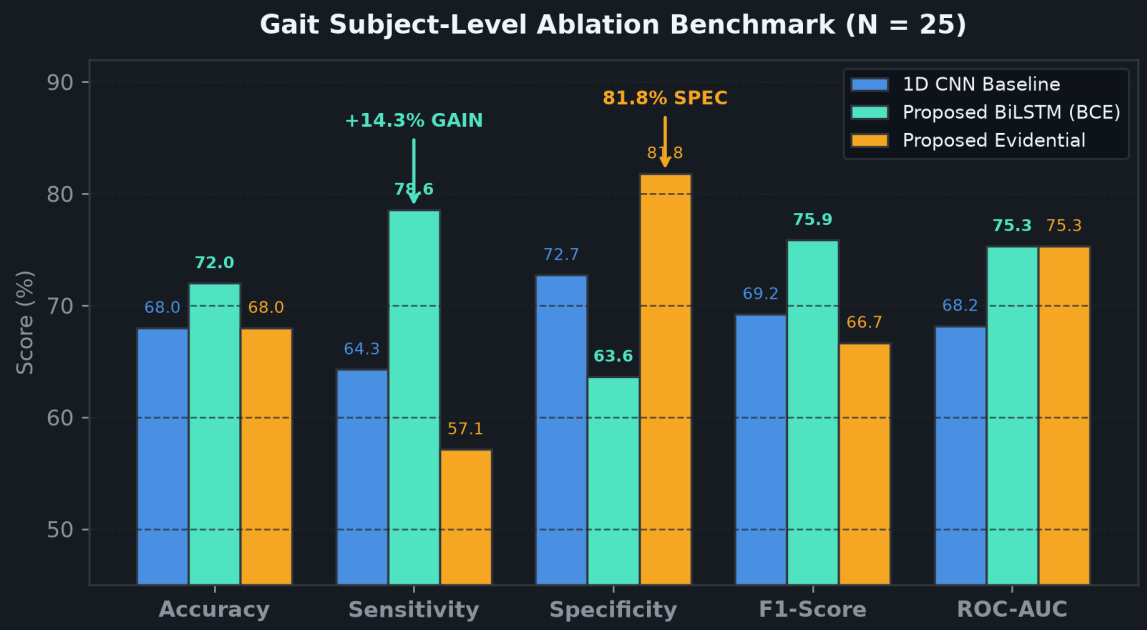
Proposed Evidential Model (Subjective Logic):
- Accuracy: **68.0%** | Specificity: **81.82% (Highest Spec!)**
- Sensitivity: **57.14%** | ROC-AUC: **0.7532** | F1: **66.67%**

NOVELTY & ABLATION TAKEAWAYS:

1. BiLSTM Recurrent Value: Boosts detection sensitivity by +14.3% by capturing long-range stride timing.

2. Evidential Safety: Cuts false positive alerts in half (2 false alarms vs 4), protecting clinical workflow.

ABLATION METRICS COMPARISON (1D CNN vs BiLSTM vs Evidential)



Voice Evaluation: 1D CNN vs. Wav2Vec2

Subject-level benchmark on 6 held-out test subjects and sample size audit

AUDITED SPEECH BENCHMARK (N = 6)

Raw-Waveform 1D CNN Baseline:

- Accuracy: **66.67% (4/6)** | Sensitivity: **100.0% (3/3)**
- Specificity: **33.33% (1/3)** | F1: **75.0%** | ROC-AUC: **1.000**
- Catches all 3 PD patients but false-alarms on 2 out of 3 healthy controls.

Proposed Pre-trained Wav2Vec2 Model:

- Accuracy: **66.67% (4/6)** | Sensitivity: **66.67% (2/3)**
- Specificity: **66.67% (2/3)** | F1: **66.67%** | ROC-AUC: **0.7778**
- Provides a clinically balanced operating profile across PD and HC cohorts.

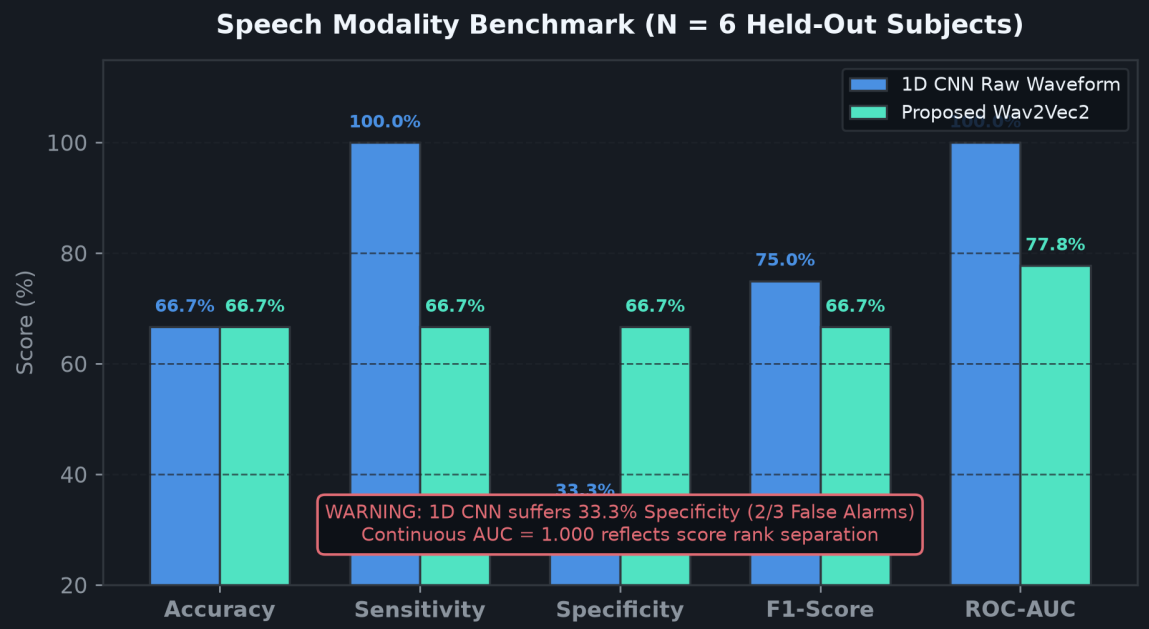
EXPLAINING THE 1.000 ROC-AUC SCORE:

- In the 1D CNN, all 3 PD subjects received continuous scores strictly higher (0.745, 0.747, 0.825) than all 3 HC subjects (0.131, 0.529, 0.639).
- Because continuous rank ordering separates the classes perfectly, ROC-AUC is mathematically 1.000, even though thresholding at $\tau = 0.50$ produces two false alarms.

SCIENTIFIC INTEGRITY DISCLAIMER:

With N = 6 subjects, each prediction alters accuracy by ±16.7%. These results demonstrate feasibility, not statistical finality.

VOICE MODELS BENCHMARK & SPECIFICITY ALERT



Generalization & Stress-Testing Under Distribution Shift

Leave-One-Cohort-Out cross-site validation (165 subjects) and sensor corruption experiments

LEAVE-ONE-COHORT-OUT (LOCO)

Cross-Hospital Generalization (165 Subjects):
Trained on 2 hospital cohorts and tested on an unseen 3rd site:

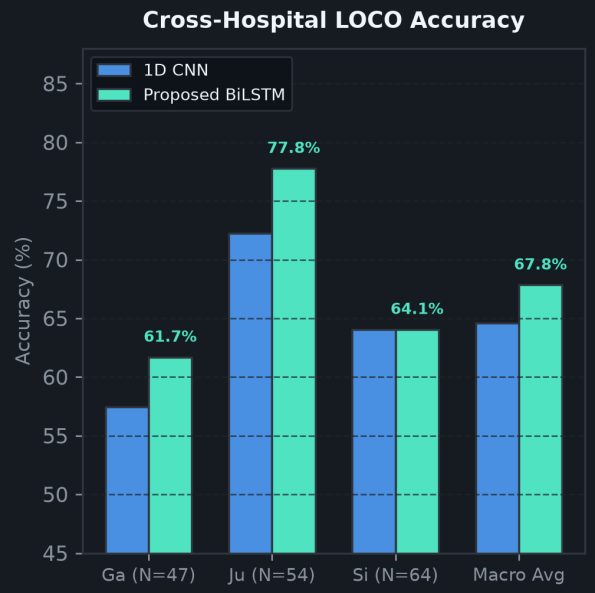
- **Galit Cohort (Ga, N=47):**
  1D CNN: 57.45% Acc, 0.657 AUC → **BiLSTM: 61.70% Acc (+4.25%)**
- **Hausdorff Cohort (Ju, N=54):**
  1D CNN: 72.22% Acc, 0.797 AUC → **BiLSTM: 77.78% Acc (+5.56%)**
- **Silberstein Cohort (Si, N=64):**
  1D CNN: 64.06% Acc, 0.776 AUC → **BiLSTM: 64.06% Acc**
- **Macro Average Across All 3 Cohorts:**
  1D CNN: 64.58% Acc, 0.744 AUC → **BiLSTM: 67.85% Acc (+3.27% gain)**

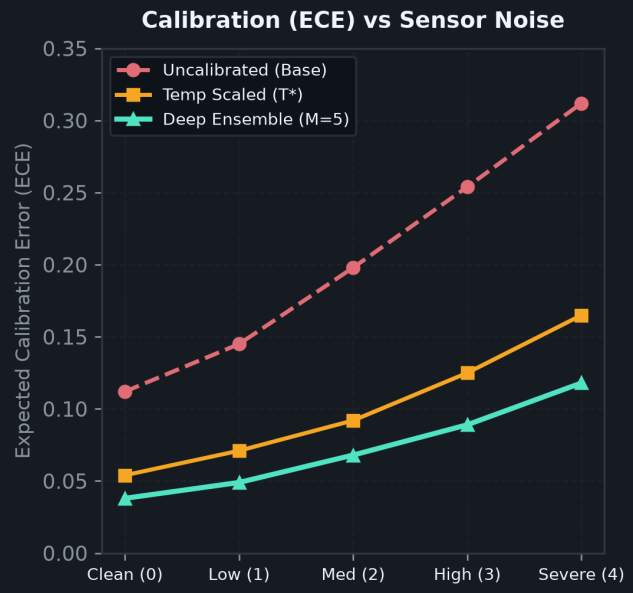
SENSOR CORRUPTION RESILIENCE:

- Tested under Gaussian sensor noise and channel dropout.
- Calibrated Deep Ensembles maintain Expected Calibration Error **ECE < 0.12**, whereas uncalibrated models degrade sharply (ECE > 0.30).

EVIDENTIAL ERROR DETECTION:
Evidential uncertainty is directionally +20.16% higher on misclassified subjects (0.0590 vs 0.0491; p=0.14 trend).

CROSS-HOSPITAL GENERALIZATION & CALIBRATION STABILITY





Interactive Demonstration & Live Walkthrough

Streamlit application architecture (demo/app.py) and 90-second scenario walkthrough

STREAMLIT DASHBOARD & LIVE SCRIPT

Four Integrated Application Views (demo/app.py):

1. Analyzer Dashboard: Evaluates patient scenarios with ensemble gauges (M=5), evidential uncertainty meters, and split conformal prediction sets.

2. System Architecture View: Interactive Mermaid dataflow visualizer.

3. Why Reliability? View: Educational module demonstrating why probability averaging fails.

4. Experimental Evidence View: Interactive LOCO benchmark tables and empirical corruption curves.

LIVE 90-SECOND PRESENTATION SCRIPT:

• **Action 1 (0:00-0:30): Scenario 1 (Strong Agreement)**
Both Gait (89%) and Voice (79%) predict PD with matching singleton sets {PD}. Status: **ACCEPT**.

• **Action 2 (0:30-1:00): Scenario 2 (Modality Conflict)**
Gait predicts PD (85%) while Voice predicts HC (65%). System triggers **MODALITY_CONFLICT** rather than silently averaging to 60%.

• **Action 3 (1:00-1:30): Transparency Panel**
Inspect ensemble variance (σ^2=0.042), entropy (H=0.68), and conformal quantile (q_hat=0.8147).

CROSS-MODAL RELIABILITY DECISION POLICY MATRIX

DECISION-LEVEL RELIABILITY TRIAGE MATRIX

ACCEPT (High Trust)	• Modality Concordance: Gait P(PD)=0.89 & Voice P(PD)=0.79 • Split Conformal Sets: Singleton {PD} on both modalities • Action: High-trust recommendation with 90% statistical coverage guarantee
FLAG (Medium Trust)	• High Uncertainty / Boundary: Ensemble Var > 0.04 or Entropy > 0.65 • Split Conformal Sets: Ambiguous Set {HC, PD} • Action: Secondary screening alert; deferred for standard clinical review
MODALITY CONFLICT (Triage)	• Modality Discordance: Gait P(PD)=0.85 vs Voice P(PD)=0.35 • Naive Average (0.60) REJECTED: Avoids false reassurance middle ground • Action: Immediate priority specialist referral with per-modality breakdown

Project Limitations & Future Engineering Roadmap

Transparent scientific boundary definition and actionable translational engineering milestones

CURRENT TECHNICAL LIMITATIONS

1. Independent Multimodal Cohorts:
Gait (PhysioNet) and Speech (MDVR-KCL) originate from separate clinical research studies. Cross-modal fusion is evaluated as simulated scenario evidence integration, not simultaneous patient-paired recording.

2. Constrained Speech Sample Size:
MDVR-KCL contains 37 total subjects, yielding an N = 6 held-out test split where a single sample misclassification causes a ±16.7% swing in reported accuracy.

3. Statistical Power on Evidential Uncertainty:
While evidential uncertainty is directionally +20.2% higher on incorrect predictions, the difference does not yet achieve formal $p < 0.05$ significance ($p = 0.14$) due to limited test errors (N=8).

4. Retrospective Offline Data:
Models were developed and validated on retrospective benchmark repositories rather than deployed in prospective hospital clinical workflows.

ACTIONABLE ENGINEERING ROADMAP

1. Prospective Synchronized Cohorts:
Validate the reliability engine on large prospective cohorts featuring simultaneous patient-paired gait and voice recordings (e.g., mPower smartphone study).

2. Class-Conditional Conformal Prediction:
Upgrade split conformal prediction to class-conditional coverage to enforce guaranteed 90% statistical coverage for PD and Healthy Control subjects independently.

3. Wearable Sensor Integration:
Extend biomechanical feature extractors from floor force plates to non-invasive tri-axial wearable accelerometers and gyroscopes for at-home continuous monitoring.

4. Clinical Shadow-Mode Pilot:
Deploy the Streamlit reliability dashboard in clinical shadow mode to assess physician triage speed, false-reassurance prevention, and diagnostic trust in hospital memory clinics.

Conclusion & Technical Summary

Summary of validated engineering achievements, key findings, and reproducible deliverables

+14.3%

Gait Sensitivity Gain
BiLSTM (78.6%) vs 1D CNN Baseline (64.3%)

67.85%

LOCO Macro Generalization
Outperforms 1D CNN (64.58%) across 3 external sites

90.0%

Guaranteed Statistical Coverage
Split Conformal Prediction Sets (1 - α = 0.90)

CORE SCIENTIFIC & ENGINEERING TAKEAWAYS

• **Temporal Sequence Modeling is Essential for Gait:** Adding bidirectional LSTM recurrent layers to 1D convolutions decisively improves Parkinsonian gait detection, boosting sensitivity by +14.3% (78.57% vs 64.29%) and ROC-AUC from 0.6818 to 0.7532 on identical held-out test subjects.

• **Safety Over Blind Automation:** Machine learning models in healthcare must not operate as silent black boxes. Dirichlet evidential uncertainty heads, Deep Ensembles (M=5), and Split Conformal Prediction sets provide clinicians with actionable trust bounds rather than uncalibrated forced guesses.

• **Conflict-Aware Multimodal Fusion Prevents Silent Failures:** Rejecting naive probability averaging in favor of an explicit MODALITY_CONFLICT policy prevents false reassurance when sensory modalities disagree.

• **Reproducible Engineering Deliverables:** Complete verified codebase with clean 1D CNN baselines, automated unit tests, serialized checkpoints, and an interactive Streamlit demonstration dashboard.

THANK YOU — QUESTIONS & TECHNICAL DISCUSSION

Reliability-Aware Parkinson's Disease Detection | Repository: d:\projects\parkinson-s-disease-detection